ECE 3441 Homework **Unit 4** **Name/ID#**______________________

4.1 – 4.9

4.1 Represent each of the following sentences by a Boolean equation.

a. The company safe should be unlocked only when Mr. Jones is in the office or Mr. Evans is in the office, and only when the company is open for business, and only when the security guard is present.

b. You should wear your overshoes if you are outside in a heavy rain and you are wearing your new suede shoes, or if your mother tells you to.

c. You should laugh at a joke if it is funny, it is in good taste, and it is not offensive to others, or if it is told in class by your professor (regardless of whether it is funny and in good taste) and it is not offensive to others.

d. The elevator door should open if the elevator is stopped, it is level with the floor, and the timer has not expired, or if the elevator is stopped, it is level with the floor, and a button is pressed.

4.2 A flow rate sensing device used on a liquid transport pipeline functions as follows. The device provides a 5-bit output where all five bits are zero if the flow rate is less than $10 \text{ gallons per minute}$. The first bit is 1 if the flow rate is at least $10 \text{ gallons per minute}$; the first and second bits are 1 if the flow rate is at least $20 \text{ gallons per minute}$; the first, second, and third bits are 1 if the flow rate is at least $30 \text{ gallons per minute}$; and so on. The five bits, represented by the logical variables A, B, C, D, and E, are used as inputs to a device that provides two outputs Y and Z.

a. Write an equation for the output Y if we want Y to be 1 iff the flow rate is less than $30 \text{ gallons per minute}$.

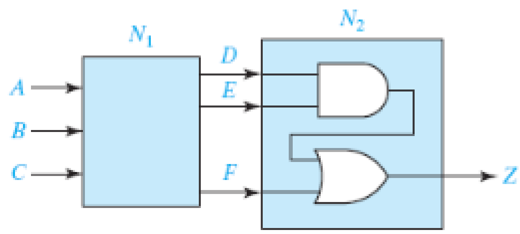
b. Write an equation for the output Z if we want Z to be 1 iff the flow rate is at least $20 \text{ gallons per minute}$ but less than $50 \text{ gallons per minute}$.

4.3 Given $F_1 = \Sigma m(0, 4, 5, 6)$ and $F_2 = \Sigma m(0, 3, 6, 7)$ find the minterm expression for $F_1 + F_2$. State a general rule for finding the expression for $F_1 + F_2$ given the minterm expansions for F_1 and F_2. Prove your answer by using the general form of the minterm expansion.

4.4 **a.** How many switching functions of two variables (x and y) are there?

b. Give each function in truth table form and in reduced algebraic form.

4.5 A combinational circuit is divided into two subcircuits N_1 and N_2 as shown. The truth table for N_1 is given. Assume that the input combinations $ABC = 110$ and $ABC = 010$ never occur. Change as many of the values of D, E, and F to don't-cares as you can without changing the value of the output Z .



A	B	C	D	E	F
0	0	0	1	1	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	1	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	0	1	0
1	1	1	0	0	0

4.6 Work (a) and (b) with the following truth table:

A	B	C	F	G
0	0	0	1	0
0	0	1	X	1
0	1	0	0	X
0	1	1	0	1
1	0	0	0	0
1	0	1	X	1
1	1	0	1	X
1	1	1	1	1

a. Find the simplest expression for F , and specify the values of the don't-cares that lead to this expression.

b. Repeat (a) for G *(Hint:* Can you make G the same as one of the inputs by properly choosing the values for the don't-care?)

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4.7 Each of three coins has two sides, heads and tails. Represent the heads or tails status of each coin by a logical variable (A for the first coin, B for the second coin, and C for the third) where the logical variable is 1 for heads and 0 for tails. Write a logic function $F(A, B, C)$ which is 1 iff exactly one of the coins is heads after a toss of the coins. Express F

a. as a minterm expansion.

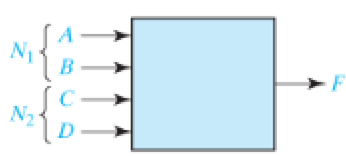
b. as a maxterm expansion.

4.8 A switching circuit has four inputs as shown. A and B represent the first and second bits of a binary number N_1 . C and D represent the first and second bits of a binary number N_2 . The output is to be 1 only if the product $N_1 \times N_2$ is less than or equal to 2 .

a. Find the minterm expansion for F .

b. Find the maxterm expansion for F .

Express your answers in both decimal notation and algebraic form.



4.9 Given: $F(a,\ b,\ c) = abc' + b'$.

a. Express F as a minterm expansion. (Use m -notation.)

b. Express F as a maxterm expansion. (Use M -notation.)

c. Express F' as a minterm expansion. (Use m -notation.)

d. Express F' as a maxterm expansion. (Use M -notation.)